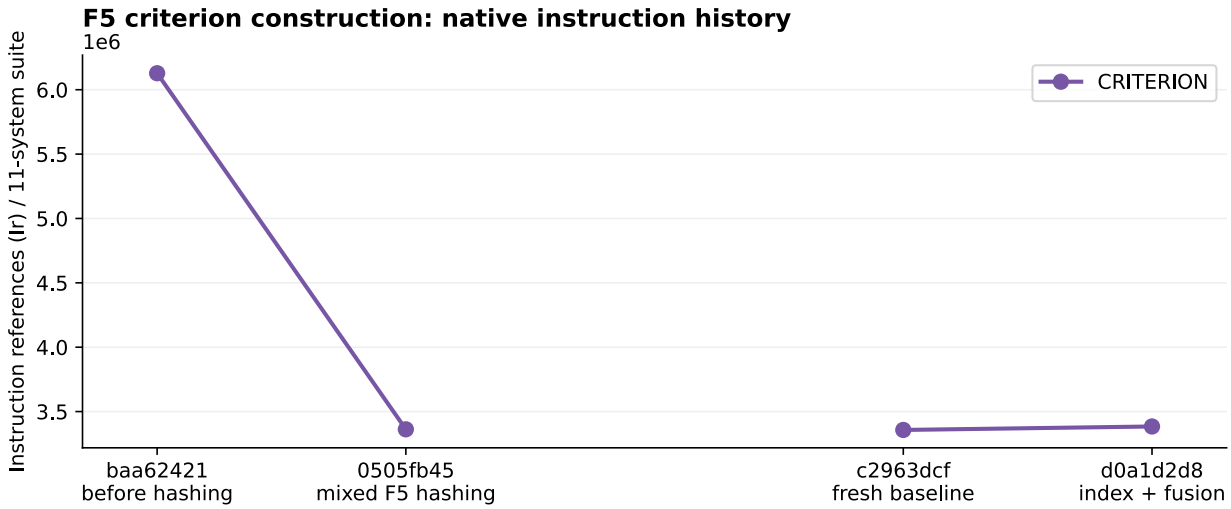
F5 criterion construction

Native instruction and allocation measurements · run-20261012T0543Z



Each segment: 5 paired rounds, 11 systems, native medians and all samples; merged workers.
run-20261012T0543Z · Gap crosses unrelated main updates; both studies' comparison.json.

CRITERION: 3,357,250 → 3,384,189 Ir per suite. Paired instruction reduction −0.80% (95% bootstrap interval −0.808% to −0.791%). Native verdict: neutral.

Criterion production source/rules are unchanged; the standalone aggregate is neutral. Three cases have more-instruction verdicts: the two n8/d3 seeds (reductions −2.31% and −5.16%) and holdout n12/d4 (−4.07%). Their cause is not established here, and the original all-arm gate fails. F5 counts full F4 rows and builds selected rows in one traversal; F4 uses a memory-bounded direct index and sorted readback terms. Pruning rules, signature order, row limits and counted XOR work are unchanged. Standalone criterion cost is a phase result; it is already included in complete F5 calls and is not added a second time.

Native sources: final comparison and raw-record hashes, frozen protocol, first rejection, bounded correction, gate scope, fixture repair, report and validation. Raw Callgrind dumps and Rust JSONL remain beside the data. Earlier segment: hashing study. Validation: 3,995 release-library tests passed, zero failed, 112 ignored; complete named coverage verified. The long ignored isogeny test passed separately. 46 Groebner/F4, 18 F5, 21 polynomial, 24 native coordinate and five comparison tests passed; Cargo examples and Clippy passed. Earlier failed/interrupted checks are retained. The inherited fixture-only repair preserves order rejection and strict parsing. Runtime and process RSS remain unmeasured.